Tallinn University of Technology

ATR0030 Robotics

Laboratory Practice Report

Lab 1 — Mitsubishi RV-2AJ

CIROS Studio / MELFA BASIC IV

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Course: ATR0030 Robotics

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1 Robot Identification

The robot used in the laboratory was the **Mitsubishi RV-2AJ**, a five-axis vertical articulated (jointed-arm) manipulator.

The robotic station was operated through Festo CIROS Studio and programmed using MELFA BASIC IV.

Table 1: Robot characteristics

Robot	Mitsubishi RV-2AJ
Degrees of freedom	5
Robot type	Vertical articulated manipulator
Joint types	Revolute (rotational)
Body names in CIROS	Base, Waist, Shoulder, Elbow, Pitch, Roll
End effector	Gripper / tool head

In the CIROS Studio Model Explorer, the RV-2AJ model is divided into the bodies **Base, Waist, Shoulder, Elbow, Pitch, and Roll**. The Base is the fixed body of the manipulator, while the robot has five rotational degrees of freedom.

The gripper was controlled in the MELFA BASIC program using the `HOPEN` and `HCLOSE` commands.

2 Functional Block Diagram of the Robotic Station

Figure 1 shows the functional relationships between the main components of the robotic station.

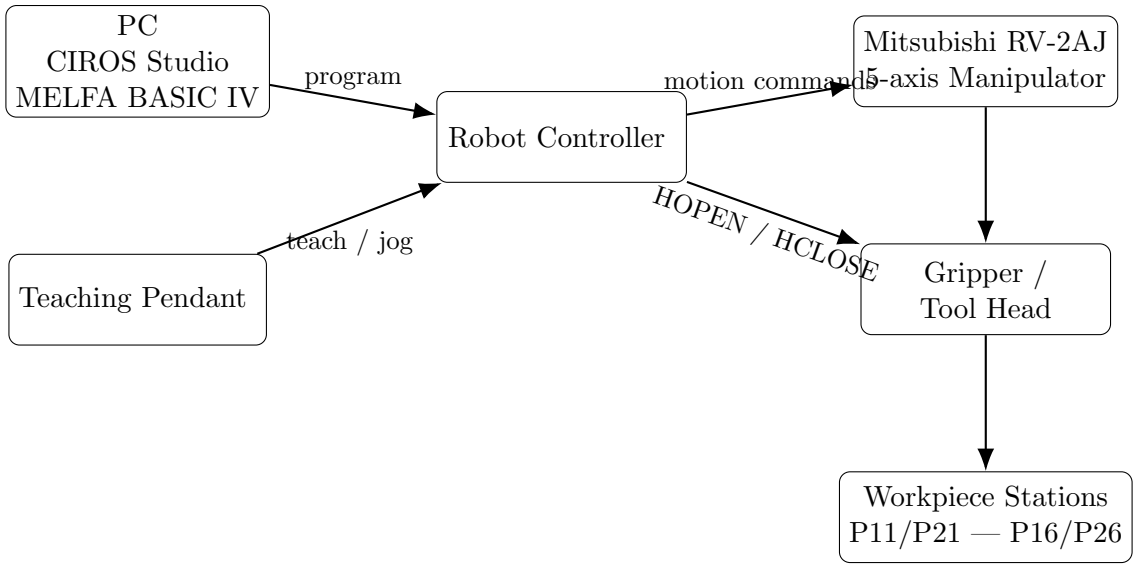


Figure 1: Functional block diagram of the Mitsubishi RV-2AJ robotic station.

The program was prepared in CIROS Studio and executed through the robot controller. But before we even created the program we used the teaching pendant to jog the robot and teach target positions. We created 8 positions all hovering over a different part the P11, P12, P14, and

P16 are the lower positions used to pick up the object with the HOpen and Hclose command on the End effector. The P21, P22, P24, and P26 positions are the positions above the lower ones. The controller commands the five robot joints and the gripper to perform the required pick-and-place sequence.

3 Taught Positions

The positions used by the program are stored as position variables. The P2x positions are approach/retract positions above the corresponding P1x working positions.

Table 2: Robot positions used during the laboratory exercise

Point	X	Y	Z	A	B	C	Role
P11	+165.98	-201.77	+278.07	-132.40	+180.74	+0.00	Pick point, station 1
P21	+165.98	-201.77	+366.23	-132.40	+180.73	+0.00	Approach point above P11
P12	+336.71	+48.94	+214.86	-132.40	+180.73	+0.00	Place/re-pick point, station 2
P22	+336.71	+48.94	+323.09	-132.40	+180.73	+0.00	Approach point above P12
P14	+369.53	+149.25	+216.38	-164.50	+180.73	+0.00	Place/re-pick point, station 3
P24	+369.53	+149.26	+290.83	-164.50	+180.72	+0.00	Approach point above P14
P16	+199.81	+177.13	+149.96	-164.50	+180.72	+0.00	**Place point, station 4**
P26	+199.82	+177.13	+326.31	-164.50	+180.72	+0.00	**Approach point above P16**

Highlighted rows indicate the positions that I personally taught during the laboratory.

4 MELFA BASIC IV Program

This is the program that we created to run the path of the robot.

The robot first approaches a working position using `MOV`. Because this movement will find the quickest way to the point. The final vertical approach and withdrawal are performed using `MVS`. The MVS causing the movement to be straight and not cut corners. The gripper is operated using `HOPEN` and `HCLOSE`.

```
1  5SPD 50'set speed to 50%
2
3  10MOV P21 'move to approach point above station 1
4  20HOPEN 1'open the tool head
5  30DLY 1'delay before next movement
6  40MVS P11 'move straight down to P11
7  50HCLOSE 1'close tool head and grasp workpiece
8  60DLY 1'delay before next movement
9  70MVS P21 'move straight back up to P21
10
11 90MOV P22 'move to approach point above station 2
12 100DLY 1'delay before next movement
13 110MVS P12 'move straight down to P12
14 120HOPEN 1'release workpiece
15 130DLY 1'delay before next movement
16 140MVS P22 'move back up to P22
17 150DLY 1'delay before next movement
18 160MVS P12 'move back down to P12
19 170HCLOSE 1're-grasp workpiece
20 180DLY 1'delay before next movement
```

```
190MVS P22 'move straight up to P22

200MOV P24 'move to approach point above station 3
210DLY 1'delay before next movement
220MVS P14 'move straight down to P14
230HOPEN 1'release workpiece
240DLY 1'delay before next movement
250MVS P24 'move back up to P24
260DLY 1'delay before next movement
270MVS P14 'move back down to P14
280HCLOSE 1're-grasp workpiece
290DLY 1'delay before next movement
300MVS P24 'move straight up to P24

310MOV P26 'move to approach point above station 4
320DLY 1'delay before next movement
330MVS P16 'move straight down to P16
340HOPEN 1'release workpiece
350DLY 1'delay before next movement
360MVS P26 'move straight back up to P26
370DLY 1'delay before next movement
380MVS P16 'move back down to P16
390HCLOSE 1're-grasp workpiece
400DLY 1'delay before next movement
410MVS P26 'move straight back up to P26

420WAIT M_IN(5) = 1'wait for external input signal 5
430END 'end program
```

5 Conclusions

Robot Teaching

Teaching the robot taught me how important precision is when defining and testing positions. Even a small error in a taught position can affect how accurately the robot places or picks up a workpiece. In our case, the final position was slightly inaccurate, which caused the workpiece to be placed at an angle instead of completely straight. The robot was still able to pick it up again, but this demonstrated how important it is to carefully teach, test, and refine each position before running the complete program.

Programming

Creating the program was very interesting.

I learned the importance of the approach and retract positions. Rather than moving directly between every working position, the robot first moves to a safe position above the workpiece and then moves straight down to interact with it this is the purpose of MVS for going up and down and MOV between the positions. This makes the motion more predictable and reduces the risk of hitting the workpiece or another part of the station.

I also learned how robot motion has to be coordinated with the gripper. Commands such as `HOPEN`, `HCLOSE`, and `DLY` allow the program to control when the gripper opens and closes and to give the mechanism enough time to complete an action before the next movement begins.

During the laboratory I also asked questions about the pneumatic components of the system. This helped me better understand how pneumatic actuators, solenoid valves, and sensors can work together with the robot controller. It was useful to connect concepts from the pneumatics class that I am in this semester with an actual robotic system rather than only studying them theoretically.

Program Execution

One important lesson from the laboratory I learned was to document a successful test as soon as it happens. After we successfully ran the program, the system later stopped working and we had to spend time troubleshooting before we could run it again. Fortunately, we were able to get the system working again. From this experience, I learned that problems can occur even after a successful test, so it is useful to record a successful run immediately and to be prepared for troubleshooting.

The completed program successfully moved the workpiece through the different stations using a repeated sequence of approach, vertical movement, gripper operation, and retraction. The slight error at the final placement also showed that successful program execution does not necessarily mean that every taught position is precise. Testing the complete sequence and observing the actual placement of the workpiece is therefore an important part of robot programming.

Overall, the laboratory was an interesting introduction to robot teaching and programming. I feel like I could create an automated system now! It helped connect robot motion, position teaching, programming, pneumatics, sensors, and troubleshooting into one working system.

6 Video

Please see the video to see how our program functions on the Mitsubishi RV-2AJ.

End of report